

TUN-YUAN CHANG (張惇媛)

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EDUCATIONS

National Tsing Hua University, Hsinchu, Taiwan Sep. 2025 - Present
Ph.D. in Information Systems and Applications GPA: 4.26/4.30
Advisor: Prof. Cheng-Hsin Hsu
Thesis: Optimizing Context-Aware Edge-Cloud AI Assistants in the MLLM Era

National Tsing Hua University, Hsinchu, Taiwan Sep. 2024 - Aug. 2025
M.S. in Information Systems and Applications
Transfer into the doctoral program with academic performance and research potential

National Tsing Hua University, Hsinchu, Taiwan Sep. 2020 - Jun. 2024
B.S. in Computer Science GPA: 3.90/4.30
Minor in Quantitative Finance

PUBLICATIONS

Journal Article

- [1] **Microservice Provisioning and Event-Driven Adaptation in Heterogeneous IoT Settings**
Yuqiao Li, Fangqi Liu, *Tun-Yuan Chang*, Cheng-Hsin Hsu, and Nalini Venkatasubramanian
Elsevier Pervasive and Mobile Computing, vol. 118, Article 102195, May 2026

Conference Papers

- [1] **DOME: Dynamic On-Premise MCP Edge Orchestration for AI Agents**
Cheng-Chia Lai, *Tun-Yuan Chang*, and Cheng-Hsin Hsu
In Proc. of ACM/IEEE Symposium on Edge Computing (SEC'26), Santa Clara, CA, October 2026
- [2] **Egocentric Daily Video Question Answering with Token-Efficient Storyboard Retrieval**
Zihao Ding, *Tun-Yuan Chang*, Cheng-Hsin Hsu, and Yao Liu
In Proc. of ACM International Workshop on Network and Operating System Support for Digital Audio and Video (NOSSDAV'26), Hong Kong, April 2026
- [3] **Enabling Ubiquitous Ambient Intelligence in the MLLM Era: From Spatio-Temporal Embeddings to Dynamic Edge Orchestration**
Tun-Yuan Chang
In Proc. of ACM Multimedia Systems Conference (MMSys'26), Hong Kong, April 2026 (Doctoral Symposium)
- [4] **Harvesting Temporal Correlation in Large Vision-Language Models: Using Pose Estimation as a Case Study**
Tun-Yuan Chang, Kenneth Chandra, and Cheng-Hsin Hsu
In Proc. of ACM International Workshop on Large Vision-Language Model Learning and Applications (LAVA'25), Dublin, Ireland, October 2025
- [5] **Resource-Constrained Lifelogging: A Utility-Based Replacement Policy for Video Understanding**
Tun-Yuan Chang and Cheng-Hsin Hsu
In Proc. of ACM Workshop on Mobile Immersive Computing, Networking, and Systems (ImmerCom'26), Austin, TX, October 2026, under review

AWARDS

- **Graduate Research Fellowship**, National Science and Technology Council of Taiwan 2025
- **Travel Grant** (ACM MM'25), ACM Special Interest Group in Multimedia (SIGMM) 2025
- **Travel Grant** (NSTC'25), National Science and Technology Council of Taiwan 2025

RESEARCH INTERESTS

Multimodal Large Language Models (MLLMs), Edge Computing, and Distributed Systems

EXPERIENCE

Ambient Intelligence for Immersive Networked Systems Lab (AIINS), National Tsing Hua University Mar. 2024 - Present

Research Assistant

PI: Prof. Cheng-Hsin Hsu

Research Description: Multimodal Large Language Models, Edge and Mobile Computing, and Intelligence Multimedia Systems

- **Context-Aware Edge-Offloaded AI Assistants**
Developing edge-offloaded systems and algorithms for multimodal large language models to support various downstream tasks across the edge-cloud continuum.
Collaborated with Prof. Yao Liu from Rutgers University and MediaTek
- **NS-3 and MAVLink-Based Unmanned Aerial Vehicle (UAV) Simulation**
Building a testbed for UAV communication in fluctuating network conditions and complex environments.
Collaborated with Prof. Nalini Venkatasubramanian from University of California, Irvine

Introduction to Computer Networks Course, National Tsing Hua University Fall 2024 & Spring 2026

Teaching Assistant

- Offering tutorials to support students in understanding fundamental concepts.
- Collaborating with course instructors in designing course materials, assignments, exams, and grading.

Center for Tactile Internet with Human-in-the-Loop (CeTI), Technische Universität Dresden Oct. 2024

Visting Researcher

- Discussing research ideas and findings with host researchers.
- Exploring opportunities for future research collaboration.

Network Research Lab, Institute of Information Science, Academic Sinica Jun. 2024 - Aug. 2024

Summer Intern

PI: Prof. Ling-Jyh Chen

Research Description: Internet-of-Things and Navigation Systems

- Developing a navigation system using the open-source project GraphHopper, supported by real-time sensor readings from IoT devices.
- Creating an interactive map visualization with the open-source project Leaflet, featuring air quality readings and route planning assistance.

Last update: August 11, 2026